

Summary chapter-2

address: X5-X7 and, X19-X27

Assembly

add: save address, r01, r02 (+)

sub: save address, r01, r02 (-)

ld: save address, 8X index [base address] (load)

sd: what we want to save, 8X index [base address] (save)

addi: save address, r01, number (a=a+4)

slli: save address, LS shift, LS shift (Left shift)

srli: save address, RS shift, RS shift (Right shift)

no index.

↓ i = 20
B = 25

A[B[i]]

- Take the address LS 2^3
- Then address + base.
- the ld.

slli: X5, X20, 3

add: X5, X5, X20

ld: X5, 0[X5]

beq: r01, r02, level 1

bgt: r01, r02, level 1

bne: r01, r02, level 1

bge: r01, r02, level 1

RISC-V to Machine:

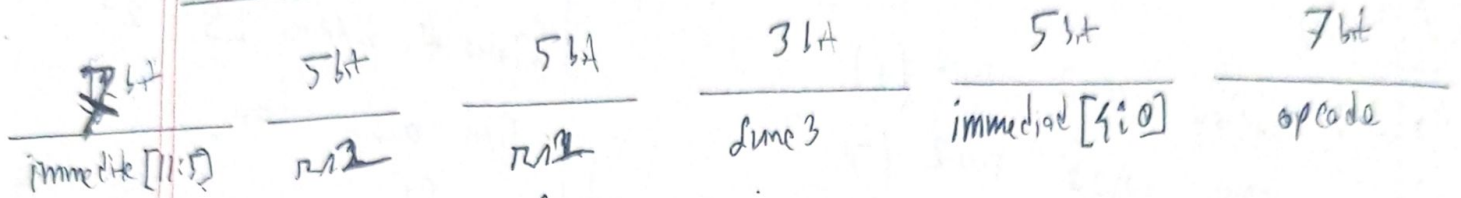
• R-type: add, sub, and, or, XOR. (convert register)

7 bit	5 bit	5 bit	3 bit	5 bit	7 bit
funct7	r02	r01 r01	funct3	rd	op code

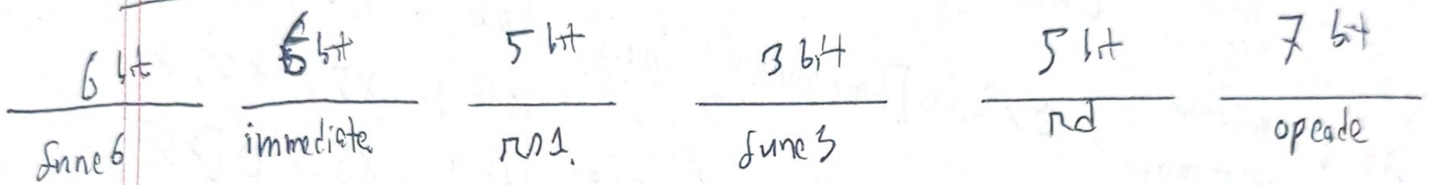
• I-type: load (convert register) (একটি r01) + destination.

12 bit	5 bit	3 bit	5 bit	7 bit
Immediate	r01	funct3	rd	op code

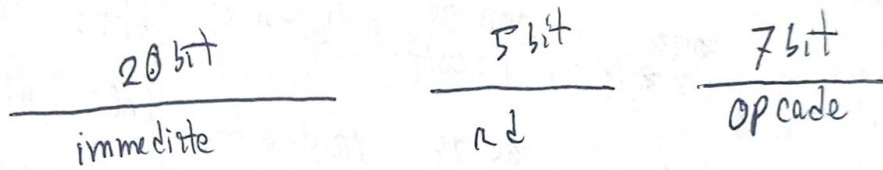
S-types rvc addren (2 source + no destination)



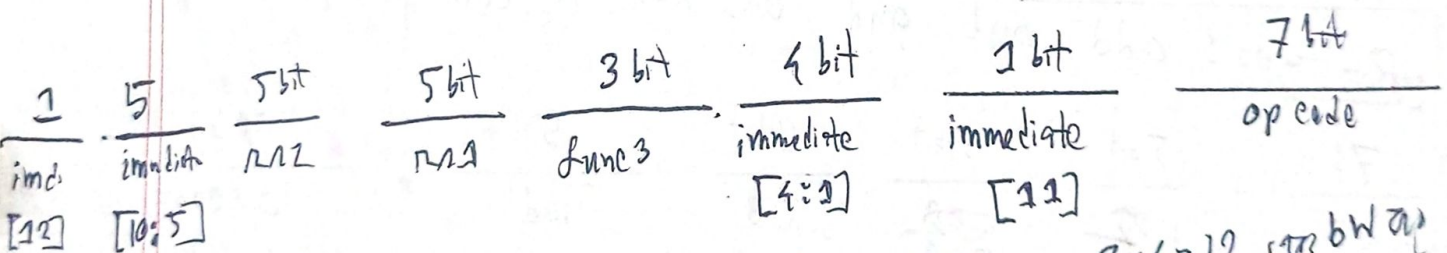
Shift-types slli, srli



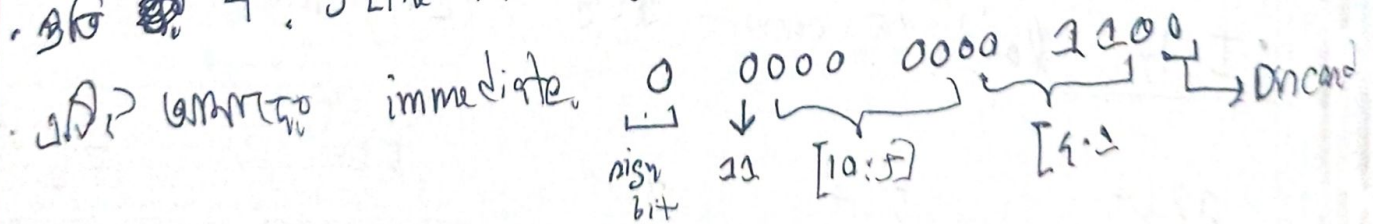
U-types lui



SB types beq, bne, blt, bge



Line 4, 3 Line Forward jump $3 \times 4 = 12$ words - 12



negativ 2's complement of 10 10

20 bit
↑
U3 type 3al

2 bit	10 bit	1 bit	8 bit	5 bit	7 bit
imm [20]	imm [10:1]	imm [11]	imm101 [19:12]	rd	op code

PC = $PC \oplus$ immediate $\times 2$

LUI: convert to ~~32 bit~~ 64 bit [lui: destination constant]

① copy 20 bit to rd. [31:12]

② Copy what you have in rest bits [63:32] according to last bit.

③ Copy zeros in [11:0] then add the copied. [11:0]